The bacterial cell envelope

THE BACTERIAL CELL ENVELOPE (sample notes)

Key components
- Cell membrane: phospholipid bilayer; site of respiration and transport in bacteria.
- Peptidoglycan (murein) cell wall: polymer of N-acetylglucosamine and N-acetylmuramic acid with peptide cross-links. The target of penicillins and cephalosporins.
- Gram-positive walls: thick peptidoglycan layer, teichoic acids.
- Gram-negative walls: thin peptidoglycan plus an outer membrane containing lipopolysaccharide (LPS, endotoxin).
- Capsule: polysaccharide layer that resists phagocytosis; a major virulence factor (e.g. Streptococcus pneumoniae).
- Flagella, pili and fimbriae: motility and attachment.

Clinical link: because LPS is part of the outer membrane of Gram-negative bacteria, it is a major driver of septic shock.